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A Project Report
on
"Freud: Mental Health AI Assistant"

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Bona fide Certificate

This project work on

“Freud: Mental Health AI Assistant”

is the bona fide work of

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Abstract

Mental health problems are increasing rapidly around the world, yet many people still do not have access to proper support. This project presents Freud: Mental Health AI Assistant, a mobile application developed for the general population to provide basic mental health support using AI. The core idea behind this project is to help middle-class individuals who are facing mental health issues but are unable to consult professional therapists due to financial limitations, as well as those who feel uncomfortable or introverted when opening up to other people.

The application uses an AI model that has been fine-tuned on custom conversational data to understand user input and respond in a supportive and empathetic manner. The model was trained after performing data collection, data parsing, and tokenization using a pre-trained base model. This allows users to express their thoughts and emotions in a private and non-judgmental environment, helping them feel comfortable and understood. The system maintains short-term conversational context and includes a basic safety mechanism to identify potential mental health risks and guide users toward professional help when necessary.

While the system does not replace professional therapy, it aims to act as an accessible and supportive companion for users who may otherwise avoid seeking help. This project highlights the potential of AI-based systems to improve mental health awareness and provide early emotional support in a simple and affordable way.

Keywords: *AI, Mental Health, Mobile Application, Chatbot, Emotional Support*

Contents

Acknowledgements **ii**

Abstract **iii**

List of Figures **vi**

List of Tables **vii**

Abbreviations **viii**

1 Introduction **1**
1.1 Background 1
1.2 Objectives 2
1.3 Motivation and Significance 3

2 Related Works **4**
2.1 Mental Health Chatbots and Conversational Agents 4
2.1.1 Commercial Systems 4
2.1.2 Academic Research Systems 4
2.2 Large Language Models and Fine-Tuning Techniques 5
2.2.1 Foundation Models 5
2.2.2 Parameter-Efficient Fine-Tuning 5
2.2.3 Conversational AI Approaches and Context Management . . 6
2.3 Privacy and Ethics in Mental Health AI 6

3 Methodology **8**
3.1 System Architecture Overview 8
3.2 Dataset Preparation and Processing 9
3.3 Model Selection and Architecture 10
3.3.1 Base Model Selection 10
3.3.2 Fine-Tuning Method 11
3.3.3 Training Configuration 11
3.3.4 Training Infrastructure 12
3.3.5 Backend Architecture 12
3.3.6 Model Deployment 12
3.3.7 Response Processing Pipeline 13
3.4 Database Architecture 14
3.4.1 Firebase Services 14
3.4.2 Database Schema 16
3.4.3 Security Rules 16
3.4.4 Privacy Implementation: TTL Cleanup 16
3.5 Frontend Development 16
3.5.1 Technology Stack 16

3.5.2 Application Architecture 17
3.5.3 Key Features Implementation 17
3.5.4 User Interface (UI)/User Experience (UX) Design 18
3.6 Integration and Testing 18
3.6.1 Integration Testing 18
3.6.2 Functional Testing 19
3.7 System Requirements Specification 19

4 Discussion on Achievement 20
4.1 Achievement Overview 20
4.1.1 Primary Accomplishments 20
4.2 Features Implemented 23
4.2.1 Features Implemented 23
4.3 Challenges and Difficulties 25
4.4 Deviations from Original Objectives 26
4.4.1 Original Objectives 26
4.4.2 Negative Deviation 27
4.4.3 Positive Deviation 27

5 Conclusion and Recommendations 28
5.1 Conclusion 28
5.2 Limitations 28
5.3 Future Enhancements 29

References 31

Appendix A 33
A.1 System Configuration Details 33
A.2 Hyperparameter Settings 33
A.3 Ethical Considerations 33
A.4 Project Timeline and Schedule 34
A.5 Screenshots 34
A.5.1 Database 34
A.6 Limitations of Supplementary Material 35

List of Figures

3.1 System Architecture Diagram 9
3.2 Backend Architecture Diagram 13
3.3 Response Processing Flowchart 14
3.4 Database Schema Diagram 15

4.1 Training and validation loss curves 21

5.1 Project Gantt Chart 34
5.2 Firebase Athentication 34
5.3 Firebase Athentication 35

List of Tables

2.1 Evolution of Mental Health AI Chatbots and Related Technologies . 7

3.1 Model Training Hyperparameters 11
3.2 System Requirements 19

Abbreviations

AI Artificial Intelligence.

API Application Programming Interface.

LLM Large Language Model.

LoRA Low-Rank Adaptation.

ML Machine Learning.

NLP Natural Language Processing.

NoSQL Not Only SQL (non-relational database).

QLoRA Quantized Low-Rank Adaptation.

SDK Software Development Kit.

TTL Time To Live.

UI User Interface.

UX User Experience.

VRAM Video Random Access Memory.

WHO World Health Organization.

Chapter 1
Introduction

This chapter introduces the Freud AI-powered mental health companion system, providing context, objectives, motivation, and expected outcomes of the project.

1.1 Background

Mental health has emerged as one of the most pressing public health challenges of the 21st century. According to the World Health Organization (WHO), approximately 280 million people worldwide suffer from depression, while anxiety disorders affect an estimated 301 million individuals (World Health Organization (2023)). In Nepal, mental health concerns are particularly acute, with limited access to professional psychiatric services, especially in rural areas, and persistent social stigma surrounding mental health treatment.

Mental health challenges have been rising worldwide, leading to a growing need for accessible and continuous psychological support. However, traditional therapy is often financially out of reach for middle-class individuals, as the cost of regular sessions can surpass what many households can reasonably afford. Moreover, people with introverted personalities or social anxiety may find face-to-face therapy intimidating, which discourages them from seeking professional help despite needing it.

Recent advances in AI, particularly in natural language processing Natural Language Processing (NLP) and Large Language Model (LLM), have opened new possibilities for mental health support. Models like GPT-3, GPT-4, and Phi-2 demonstrate remarkable conversational capabilities and can be fine-tuned for domain-specific applications. Fine-tuning techniques such as Quantized Low-Rank Adaptation (QLoRA) enable training large models on consumer-grade hardware, democratizing AI development.

However, existing mental health chatbots often suffer from generic responses, limited contextual understanding, and inability to provide genuinely empathetic support. Commercial solutions may also raise privacy concerns by storing sensitive mental health data on third-party servers. There exists a need for an accessible,

privacy-conscious, and contextually aware AI assistant specifically designed for mental health support.

The primary problems this project addresses are:

1. Limited access to mental health support: Many people cannot get professional help because therapy is expensive or mental health services are not easily available in their area.

2. Lack of mental health awareness: Many individuals do not fully understand mental health issues or where to seek help, which prevents early support and care.

3. Difficulty expressing emotions openly: Some people find it hard to talk about their feelings face-to-face and prefer a private, non-judgmental space to express themselves.

4. Unhelpful AI chatbots: Many existing chatbots give generic and robotic responses that do not feel empathetic or supportive.

5. Privacy concerns: Some mental health applications store sensitive conversations for long periods, making users worried about the safety of their personal data

1.2 Objectives

The primary objectives of the Freud AI project are:

1. To design and develop a fine-tuned conversational AI model capable of engaging in empathetic and context-aware conversations related to mental health support.

2. To build a privacy-focused mobile application that provides a safe, non-judgmental environment for users, ensuring secure data handling and automatic deletion of sensitive information.

3. To implement a safety and support mechanism that can detect signs of emotional distress and provide appropriate guidance, coping strategies, and redirection to professional support when necessary, without offering clinical diagnoses.

4. To demonstrate the feasibility and accessibility of AI-based mental health assistance by deploying large language models on resource-constrained infrastructure, making support available to users who face financial, social, or accessibility barriers.

1.3 Motivation and Significance

The motivation for this project arises from the growing mental health challenges observed among students, young professionals, and the general population. Many individuals experience stress, anxiety, and emotional distress but hesitate to seek professional help due to financial limitations, hesitation in opening up to others, or uncertainty about where to begin. These challenges are especially evident among introverted individuals who find face-to-face conversations difficult. Recognizing this gap, the project aims to provide an accessible, private, and empathetic AI-based mental health assistant that users can interact with at any time. From a technical perspective, the project is also motivated by the opportunity to explore modern AI techniques, particularly the fine-tuning of large language models (LLM) for emotionally sensitive and socially impactful applications.

The significance of this project lies in its potential social, technical, and educational impact. By offering a 24/7 AI-powered mental health assistant through a mobile application, the system improves accessibility to emotional support without financial or geographical barriers. Its privacy-focused design encourages users to engage openly in a judgment-free environment, helping reduce hesitation around discussing mental health concerns. Technically, the project shows how a large language model (LLM) can be effectively fine-tuned and deployed on limited hardware, demonstrating that real-world AI applications are possible even in resource-constrained environments. Overall, the project contributes to the growing field of ML for social good by showing how advanced ML techniques can be responsibly applied to support mental well-being.

Chapter 2

Related Works

This chapter examines existing research and systems in mental health AI, conversational agents, and large language model fine-tuning, identifying gaps that motivated the development of Freud.

2.1 Mental Health Chatbots and Conversational Agents

2.1.1 Commercial Systems

Several commercial mental health chatbots have gained prominence in recent years.

Woebot, developed by Woebot Health, applies cognitive behavioral therapy (CBT) in a conversational format and has demonstrated effectiveness in reducing symptoms of depression and anxiety in clinical studies Fitzpatrick et al. (2017). However, its subscription-based model limits accessibility for economically disadvantaged users.

Replika, originally designed as a companionship AI, has expanded to include mental health support features. Despite its advanced transformer-based conversational capabilities, concerns regarding data privacy and commercialization of user interactions have been raised Skjuve et al. (2021).

Wysa is an AI-powered mental health application that combines rule-based dialogue with limited NLP. Studies indicate that Wysa can reduce depressive symptoms Inkster et al. (2018), though its reliance on predefined conversation flows restricts conversational flexibility.

Overall, these commercial systems demonstrate the potential of AI-based mental health support but commonly face limitations related to cost, privacy, and rigid or template-driven interactions.

2.1.2 Academic Research Systems

Academic research has explored multiple approaches to conversational systems for mental health support.

Tess, developed by X2AI, applies psychological AI to deliver cognitive behavioral therapy and psychoeducation via text messaging Fulmer et al. (2018). Evaluation studies report reductions in anxiety and depression symptoms.

SimSensei and Ellie, developed by USC's Institute for Creative Technologies, employ multimodal sensing to identify indicators of psychological distress during conversations DeVault et al. (2014). These systems integrate NLP with facial expression analysis, though multimodal complexity poses deployment challenges.

Sharma et al. introduced a counseling conversation dataset and showed that transformer models fine-tuned on therapy transcripts generate more empathetic responses than general-purpose models Sharma et al. (2020). This supports the effectiveness of domain-specific fine-tuning for mental health applications.

2.2 Large Language Models and Fine-Tuning Techniques

2.2.1 Foundation Models

The advent of large language models (LLM) has significantly advanced NLP.

GPT-3 Brown et al. (2020), with 175 billion parameters, demonstrated strong few-shot learning capabilities and high-quality text generation across diverse domains. However, its large model size and proprietary nature limit accessibility for researchers and developers.

GPT-Neo and GPT-J, developed by EleutherAI, introduced open-source alternatives ranging from 125 million to 6 billion parameters Black et al. (2021). These models enabled experimentation on consumer hardware, though smaller variants often struggle with complex tasks.

Microsoft Phi-2, released in late 2023, represents a new class of compact language models Microsoft Research (2023). With 2.7 billion parameters trained on high-quality data, Phi-2 achieves performance comparable to significantly larger models, making it well-suited for fine-tuning on limited hardware.

2.2.2 Parameter-Efficient Fine-Tuning

Traditional fine-tuning of large language models updates all parameters, resulting in high computational and memory costs. Low-Rank Adaptation (LoRA) Hu et al.

(2021) addresses this by training small adapter matrices within transformer layers while keeping the base model frozen, reducing trainable parameters by orders of magnitude without sacrificing performance.

QLoRA (Quantized Low-Rank Adaptation) Dettmers et al. (2023) extends LoRA by applying 4-bit quantization to the base model, enabling fine-tuning of very large models on limited hardware. In our project, QLoRA allows fine-tuning Phi-2 (2.7B parameters) on a 16GB Kaggle P100 GPU, making large-model fine-tuning more accessible.

Dettmers et al. show that QLoRA achieves performance comparable to full fine-tuning while reducing memory usage by approximately 75% Dettmers et al. (2023). This supports our architectural decision to use QLoRA for training Freud's model.

2.2.3 Conversational AI Approaches and Context Management

Early conversational systems relied on retrieval-based methods, selecting predefined responses based on input matching Wu et al. (2017). While computationally efficient, these systems cannot generate novel responses. Generative models, using sequence-to-sequence and transformer architectures, produce original responses by learning language patterns from data Vinyals and Le (2015), providing more natural and context-aware conversations. Hybrid approaches combine retrieval and generation to ground responses in factual knowledge while allowing flexible language formulation Ghazvininejad et al. (2018). Freud adopts a purely generative design, fine-tuned to embed mental health knowledge directly into model parameters.

Maintaining context across multiple turns is crucial for coherent dialogue. Transformers handle context via self-attention but are limited by fixed context windows Vaswani et al. (2017). Techniques such as sliding window attention and memory networks have been proposed to extend context capacity Beltagy et al. (2020), enabling multi-turn conversational coherence.

2.3 Privacy and Ethics in Mental Health AI

Privacy is a critical concern in mental health applications. Balsa et al. highlight the risk of re-identification even in anonymized mental health datasets Balsa et al. (2018), while the European Union's GDPR enforces strict protections for health-related data European Parliament and Council (2016).

Our system addresses these concerns through:

- Automatic data deletion using a 90-day time-to-live (Time To Live (TTL)) policy
- Local Firebase authentication without third-party data sharing
- User-controlled conversation history management
- Clear disclosure that Freud is not a medical device

Ethical guidelines for mental health AI emphasize transparency, user autonomy, and clearly defined system boundaries Martinez-Martin and Kreitmair (2020). Freud follows these principles by explicitly communicating its limitations and encouraging professional help when serious mental health concerns arise.

Year	**System/Technology**	**Key Contribution**
2014	SimSensei & Ellie	Multimodal sensing system detecting psychological distress through facial expressions and natural language
2017	Woebot	First CBT-based conversational agent demonstrating clinical effectiveness in reducing depression and anxiety symptoms
2018	Wysa	AI-powered mental health app combining rule-based flows with limited NLP, clinically validated
2018	Tess (X2AI)	Psychological AI delivering CBT and psychoeducation via text messaging
2019	Replika	Advanced companionship AI using transformer architectures, evolved to include mental health features

Table 2.1: Evolution of Mental Health AI Chatbots and Related Technologies

Chapter 3

Methodology

This chapter provides a detailed description of the system architecture, dataset preparation, model fine-tuning, backend development, database design, and frontend implementation for Freud, an AI mental health companion.

3.1 System Architecture Overview

Freud employs a three-tier architecture consisting of:

1. Frontend Layer: Flutter-based mobile application providing user UI for authentication, conversation management, and mood tracking.
2. Backend Layer: FastAPI service deployed on Hugging Face Spaces, hosting the fine-tuned GPT-Neo 125M model and handling inference requests.
3. Data Layer: Firebase services including Firestore for conversation storage and Firebase Authentication for user management.

The system workflow operates as follows:

1. User authenticates through Firebase Authentication
2. User sends message through Flutter mobile app
3. Frontend retrieves recent conversation context from Firestore
4. Frontend sends user message and context to FastAPI backend
5. Backend performs inference using fine-tuned GPT-Neo 125M model
6. Generated response returns to frontend
7. Both user message and AI response persist to Firestore
8. TTL cleanup function automatically deletes data after 90 days

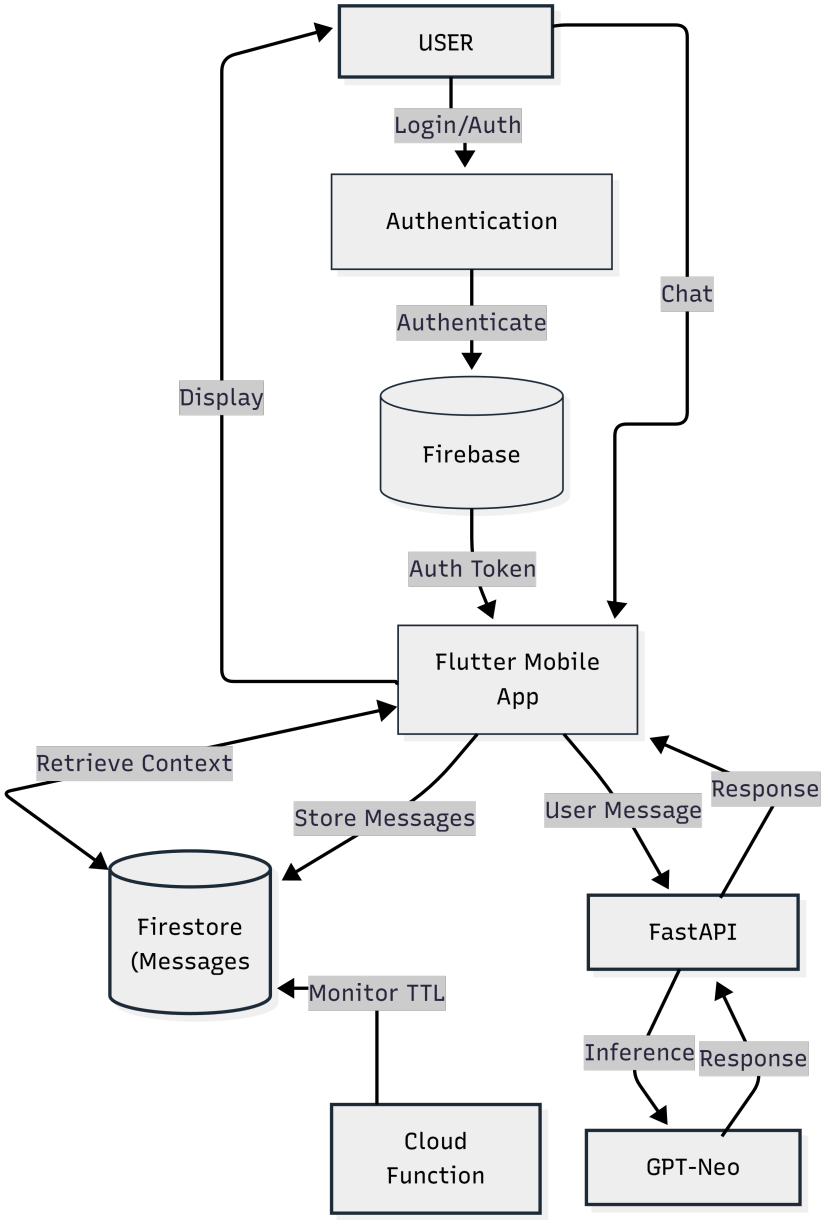

Figure 3.1: System Architecture Diagram

3.2 Dataset Preparation and Processing

The dataset used for fine-tuning the model was collected from Kaggle and consists of 11,139 mental health related conversation samples grouped into 80 different emotional intent categories. These categories include basic interactions such as greetings and gratitude, emotional states like sadness, anxiety, depression, stress, and happiness, crisis-related topics such as suicidal thoughts and self-harm, support mechanisms including meditation and mental health education, and casual conver-

sations.

The dataset contains both single-turn and multi-turn conversations, with approximately equal distribution between the two. A significant portion of the data represents negative emotional states, reflecting its focus on mental health and crisis support, while positive and neutral interactions are included to maintain natural conversational flow.

Data preprocessing involved several steps to improve quality and consistency. All conversations were converted into a standardized JSON format, text inconsistencies and encoding errors were corrected, and duplicate samples were removed to avoid overfitting. Profanity in user inputs was retained for realistic training, while assistant responses were kept professional and supportive. Conversations were structured using special tokens to clearly separate system instructions, user inputs, and assistant responses.

To improve model generalization, data augmentation techniques such as response variation, context variation, and multi-turn expansion were applied. The dataset was split into 90% training and 10% validation sets using stratified sampling to ensure balanced emotion representation.

3.3 Model Selection and Architecture

3.3.1 Base Model Selection

GPT-Neo Implementation

Initially, GPT-Neo 125M by EleutherAI was selected as the base model for fine-tuning. GPT-Neo 125M is a transformer-based, open-source language model with 125 million parameters. It was chosen due to its suitability for limited hardware environments and ease of deployment.

The key reasons for selecting GPT-Neo 125M were:

1. Hardware Compatibility: The small model size allows training and inference on GPUs with 8–16GB Video Random Access Memory (VRAM).

2. Training Efficiency: Faster fine-tuning enables quicker experimentation.

3. Open-Source Support: Easily accessible through Hugging Face with strong community documentation.

4. Context Handling: Supports up to 2048 tokens, sufficient for multi-turn conversations.

5. Low Latency: Enables real-time responses during deployment.

Phi-2 Experimentation

During later stages, Microsoft Phi-2 (2.7B parameters) was explored due to its improved reasoning and response quality. Initial tests showed better conversational coherence. However, deployment on free-tier CPU infrastructure resulted in high response latency, making it unsuitable for real-time use at that stage.

As a result, GPT-Neo 125M was initially used, while Phi-2 was planned for future deployment with optimization and GPU support.

3.3.2 Fine-Tuning Method

Since GPT-Neo 125M is a relatively small model, full fine-tuning was applied instead of parameter-efficient methods such as LoRA. Full fine-tuning updates all model parameters, allowing better adaptation to mental health–focused conversations.

3.3.3 Training Configuration

Training hyperparameters were carefully selected based on hardware constraints and empirical testing:

Parameter	**Value**
Base Model	EleutherAI/GPT-Neo-125M
Fine-Tuning Method	Full Fine-Tuning
Learning Rate	5e-5
Batch Size	2
Gradient Accumulation Steps	8
Maximum Sequence Length	512 tokens
Number of Epochs	3
Optimizer	AdamW
Learning Rate Scheduler	Linear with warmup
Warmup Steps	500

Table 3.1: Model Training Hyperparameters

3.3.4 Training Infrastructure

Training was performed on Kaggle cloud infrastructure using:

- GPU: NVIDIA Tesla P100 (16GB VRAM)
- Training Duration: Approximately 30-45 min for 3 epochs
- Checkpoint Frequency: Every 500 steps

The training process included loading the model, tokenizing data, training with checkpoints, validating performance, and uploading the final model to Hugging Face for deployment.

3.3.5 Backend Architecture

The backend was developed using FastAPI, a lightweight Python framework for building Application Programming Interface (API)s. It acts as a bridge between the mobile application and the AI model.

The backend performs the following tasks:

1. Loads the fine-tuned language model
2. Receives user messages from the mobile app
3. Generates AI responses and sends them back to the frontend

3.3.6 Model Deployment

The fine-tuned model was deployed using Hugging Face Spaces, which provides free cloud hosting for machine learning applications. The deployment process included uploading the trained model to the Hugging Face Hub, configuring the required Python dependencies, and setting up an API service. The application was deployed primarily on CPU infrastructure, with optional GPU support for faster inference when available. An API endpoint was exposed to allow the mobile application to send requests and receive responses.

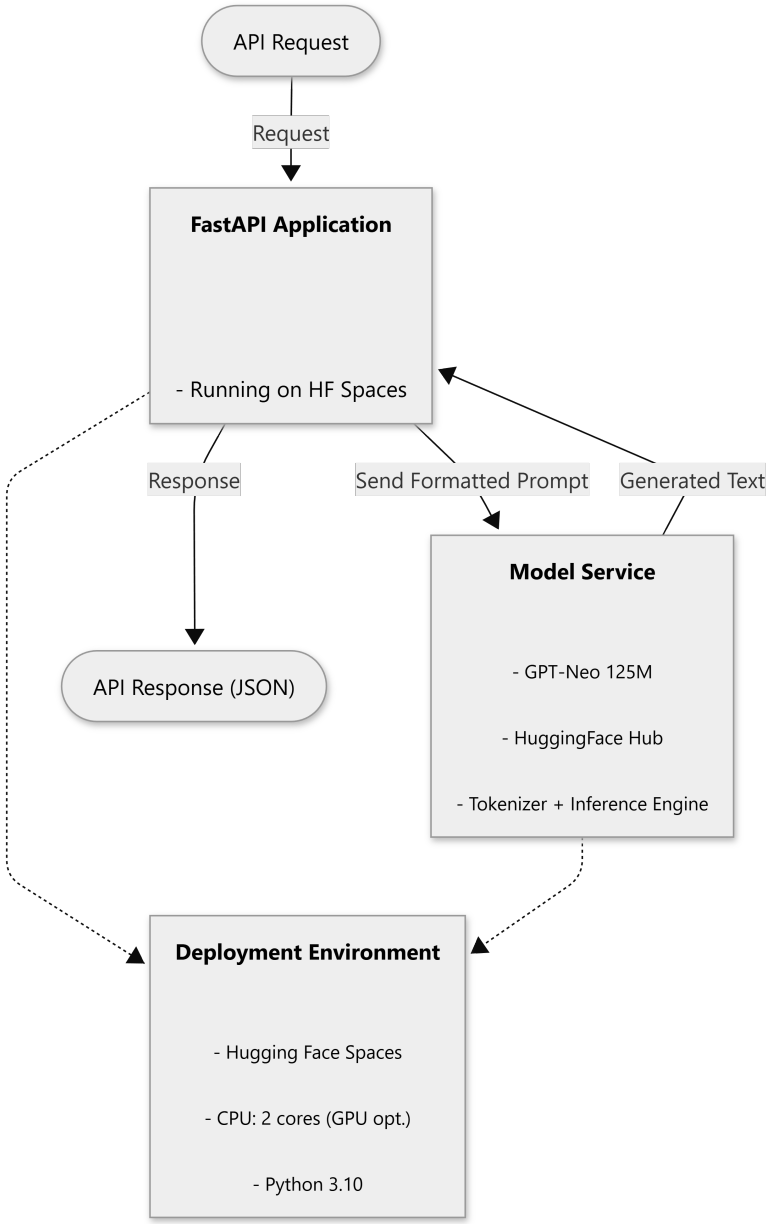

Figure 3.2: Backend Architecture Diagram

Deployment URL: https://huggingface.co/spaces/Dalton-Khatri/freud-api

3.3.7 Response Processing Pipeline

Generated responses undergo cleaning to remove artifacts:

1. Token Removal: Strip special tokens like `<|user|>`, `<|assistant|>`
2. Length Truncation: Ensure response fits within 150 tokens
3. Formatting Cleanup: Remove excessive newlines or whitespace

4. Safety Check: Detect crisis keywords and append professional help resources

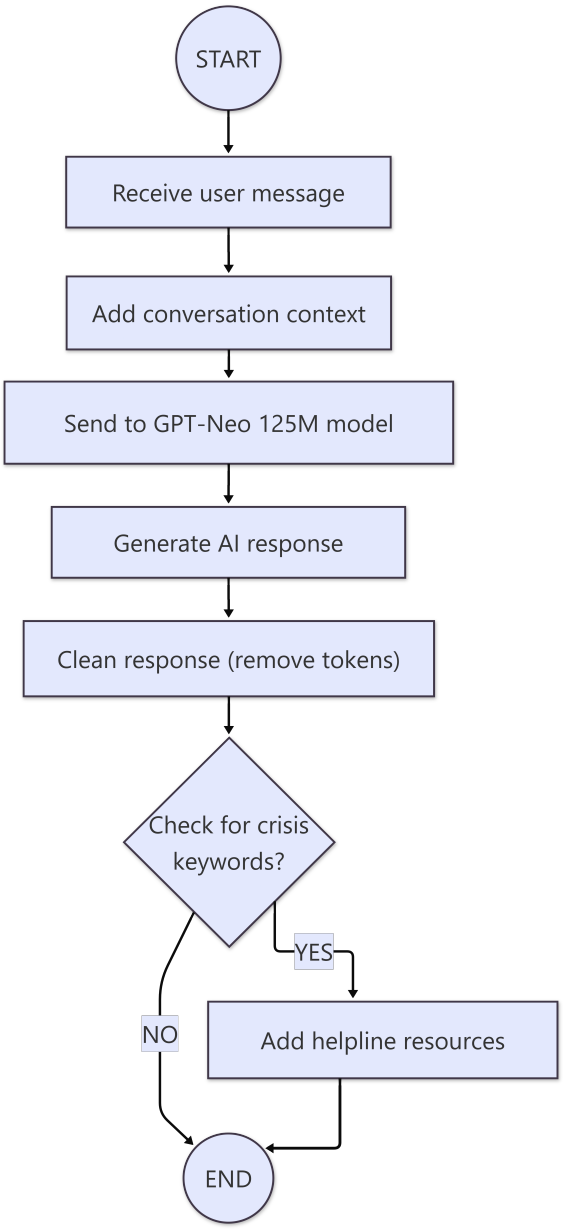

Figure 3.3: Response Processing Flowchart

3.4 Database Architecture

3.4.1 Firebase Services

The application uses Google Firebase to manage authentication and data storage.

Firebase Authentication

Handles user account management with:

- Email/password authentication
- Secure token-based session management
- Password reset functionality
- Account deletion with cascade data cleanup

Cloud Firestore

NoSQL (Not Only SQL (non-relational database) (NoSQL)) document database storing:

- User profiles
- Conversation metadata
- Message history
- Mood tracking entries
- User preferences

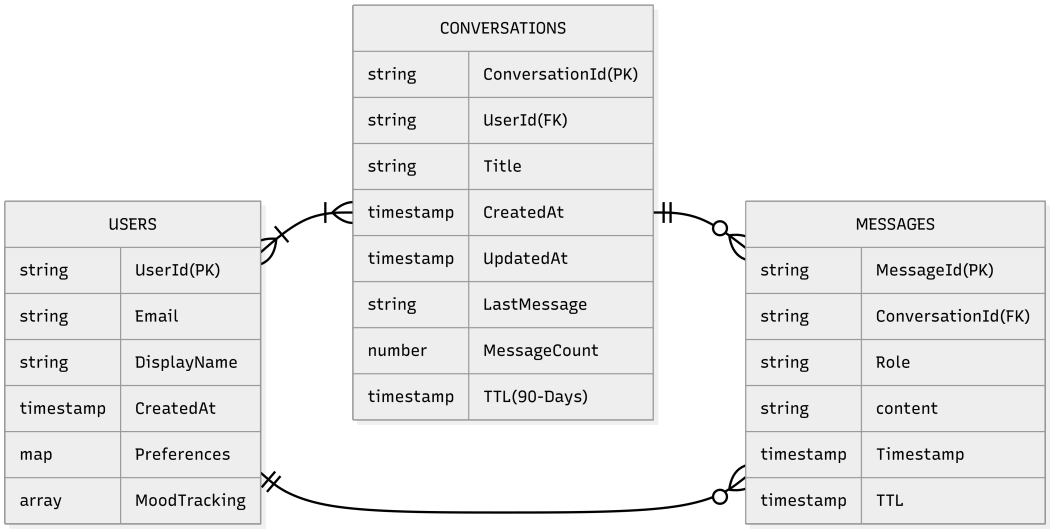

Figure 3.4: Database Schema Diagram

3.4.2 Database Schema

The Firestore NoSQL database is organized into three main collections:

1. Users Collection: Stores basic user information, preferences, and optional mood tracking data.
2. Conversations Collection: Stores conversation metadata such as title, timestamps, message count, and last message.
3. Messages Collection: Stores individual messages linked to conversations, including sender role and timestamp.

This structure allows efficient retrieval of conversations and message history while maintaining scalability.

3.4.3 Security Rules

Firestore security rules enforce data access control, ensuring users can only access their own data. Rules are configured to prevent unauthorized read or write operations across all collections.

3.4.4 Privacy Implementation: TTL Cleanup

To protect user privacy, conversation and message data are automatically deleted after 90 days. Each record includes a TTL (Time-To-Live) field, and a scheduled Firebase Cloud Function removes expired data daily. This approach also helps reduce storage usage.

3.5 Frontend Development

3.5.1 Technology Stack

The mobile application was developed using:

- Framework: Flutter 3.38.5 (Dart language)
- State Management: Provider pattern
- Authentication: Firebase Auth Software Development Kit (SDK)
- Database: Cloud Firestore SDK

- HTTP Client: Dart http package for API calls
- UI Components: Material Design 3

3.5.2 Application Architecture

The Flutter app follows a modular architecture with clear separation of concerns:

- Models: Data structures for users, conversations, and messages
- Screens: Authentication, home, chat, and profile interfaces
- Services: Firebase integration and AI API communication
- Widgets: Reusable UI components
- Utils: Constants, themes, and helper functions

3.5.3 Key Features Implementation

Authentication Flow

1. User enters email and password
2. Firebase Authentication validates credentials
3. Upon success, user profile created/fetched from Firestore
4. Authentication state persists across app sessions
5. Logout clears local session data

Conversation Management

1. Home screen displays list of user's conversations
2. Conversations sorted by updatedAt (most recent first)
3. Tapping conversation navigates to chat screen
4. Floating action button creates new conversation
5. Swipe-to-delete removes conversation and associated messages

Message Flow

1. User types message in chat input field
2. Message persists to Firestore with role "user"
3. Recent messages (last 10) retrieved as context
4. Context and new message sent to backend API
5. Backend returns AI-generated response
6. AI response persists to Firestore with role "assistant"
7. Both messages display in chat UI via StreamBuilder

3.5.4 UI/UX Design

The interface employs:

- Color Scheme: Calming purple primary color (#6C63FF)
- Typography: Google Fonts (Poppins) for accessibility
- Message Bubbles: Distinct colors for user vs. assistant
- Animations: Subtle fade-in animations for messages
- Loading States: Typing indicators during AI response generation

3.6 Integration and Testing

3.6.1 Integration Testing

Manual integration testing verified:

1. Authentication: User signup, login, logout functionality
2. Data Persistence: Messages correctly saved to Firestore
3. API Communication: Successful requests/responses with backend
4. Context Handling: Conversation history properly formatted
5. Error Handling: Graceful degradation on network failures

3.6.2 Functional Testing

Each component was tested individually:

- Firebase Service: CRUD operations on Firestore
- AI Service: API request formatting and response parsing
- Authentication: Email validation, password requirements
- UI Components: Rendering, user interactions, state updates

3.7 System Requirements Specification

This section outlines the software and hardware requirements necessary for the development and deployment of the Freud AI mental health companion system.

Software Requirements	**Hardware Requirements**
Development Environment	
Languages: Dart, Python Mobile Framework: Flutter 3.38.5 Backend: FastAPI ML Libraries: Transformers, PyTorch Database: Firebase (Firestore, Auth)	Training Infrastructure: • NVIDIA Tesla P100 GPU • 16 GB VRAM • 30 GB RAM • 73 GB Storage
Deployment Requirements	
Model Hosting: Hugging Face Spaces Database: Firebase Cloud API: RESTful (FastAPI)	Server: • CPU-based (free tier) • Optional: T4 GPU upgrade
User/Client Requirements	
Platform: Android (8.0+) Development Tools: Android Studio, VS Code, Git, Jupyter Notebook OS Support: Windows/macOS (Dev), Android (Production)	User Device: • Android smartphone • Minimum 2 GB RAM • Internet connectivity • GPS, Camera (optional)

Table 3.2: System Requirements

Chapter 4

Discussion on Achievement

This chapter presents a comprehensive analysis of what the Freud AI project achieved, the features implemented, challenges encountered, and how the final system compares to the original objectives.

4.1 Achievement Overview

The Freud AI project successfully developed a functional mental health companion application that provides empathetic conversational support to users. Despite being a student project with limited resources and a compressed three-month timeline, the team achieved the core objective of creating an accessible, privacy-focused AI chatbot for emotional support.

4.1.1 Primary Accomplishments

Functional AI Chatbot System

We successfully fine-tuned and deployed a conversational AI model (GPT-Neo 125M) capable of engaging in mental health conversations. The system demonstrates:

- Basic understanding of emotional expressions
- Context-aware response generation using recent conversation history
- Empathetic communication tone
- Ability to handle diverse emotional states (sadness, anxiety, stress, etc.)
- 24/7 availability via cloud deployment

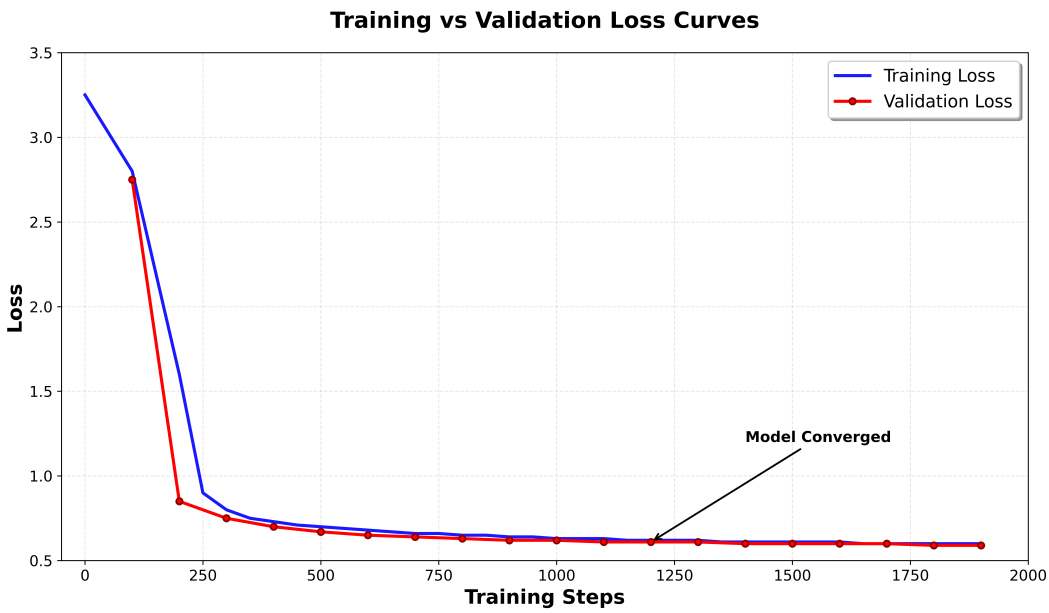

Figure 4.1: Training and validation loss curves .

Figure 4.1 demonstrates the training process across 3 epochs. The training loss (blue) decreased steadily from 3.2 to 0.6, while validation loss (red) closely followed the same trajectory, stabilizing at 0.59. The minimal gap between training and validation curves indicates the model learned generalizable patterns rather than memorizing training data, confirming successful fine-tuning without overfitting.

While the model has limitations in response quality and consistency, it successfully validates the feasibility of using fine-tuned LLMs for mental health support applications.

Complete Mobile Application

Developed a full-featured mobile application using Flutter that provides:

- Secure user authentication with email/password
- Intuitive conversation management interface
- Real-time chat functionality with message synchronization
- Basic mood tracking capabilities
- Clean, calming UI design
- Responsive performance on Android devices

The application successfully integrates frontend, backend, and database components into a cohesive user experience (UX).

Privacy-First Architecture

Implemented robust privacy protections addressing user concerns:

- Automatic 90-day data deletion via TTL mechanism
- Strong Firestore security rules preventing unauthorized access
- User-controlled conversation management
- No third-party data sharing or analytics
- Clear privacy disclaimers in application
- Immutable message history (cannot be edited after sending)

Cloud-Based Deployment

Successfully deployed the complete system on free cloud infrastructure:

- Backend hosted on Hugging Face Spaces
- Database managed by Firebase (NoSQL)
- Scalable architecture supporting multiple concurrent users
- 24/7 availability with minimal downtime
- No ongoing infrastructure costs

This achievement demonstrates that meaningful AI applications can be built and deployed without significant financial resources.

Team Learning and Growth

The project provided invaluable learning experiences for all team members:

- First exposure to transformer models and fine-tuning techniques (ML, LoRA, QLoRA)
- Practical experience with modern mobile development (Flutter)

- Understanding of cloud architecture and deployment
- Experience with NoSQL databases and real-time data
- Practice in collaborative software development
- Project management and time allocation skills

Team members progressed from having minimal AI/ML knowledge to successfully training and deploying a language model application.

4.2 Features Implemented

4.2.1 Features Implemented

User Authentication System

Implemented features:

- Email/password registration with validation
- Secure login with Firebase Authentication (SDK)
- Session persistence across app restarts

The authentication system provides enterprise-grade security through Firebase's proven infrastructure.

Conversational AI Interface

Implemented features:

- Real-time chat interface with instant updates
- Distinct message bubbles for user vs. AI messages
- Typing indicators during AI response generation
- Smooth auto-scrolling to latest messages
- Message timestamps for context
- Automatic UI updates via Firestore streams
- Conversation context maintenance (10 messages)

The chat interface provides a natural, familiar user experience similar to popular messaging applications (UX).

Conversation Management

Implemented features:

- Naming Conversation
- View conversation history sorted by recent activity
- Delete conversations (with confirmation)
- Conversation preview showing last message
- Rating the conversation
- Feedback input

Users can easily manage multiple conversation threads, returning to previous discussions when needed.

AI Response Generation

Implemented features:

- Context-aware responses using last 10 messages
- Empathetic communication tone
- Response length control (typically 50-150 words)
- Basic safety mechanisms (keyword detection)
- Fallback responses for API failures

Data Privacy and Security

Implemented features:

- Firestore security rules enforcement
- User data isolation (users cannot access others' data)
- Automatic TTL deletion after 90 days
- Immutable message history
- HTTPS encryption for all communications

The security implementation follows industry best practices and provides strong privacy guarantees.

Crisis Detection

Currently implemented:

- Basic keyword detection (e.g., ”suicide”, ”kill myself”, ”end it all”)
- Generic safety message response with helpline phone numbers
- Encouragement to seek professional help

4.3 Challenges and Difficulties

ML Learning Curve

The team’s first experience with transformer models and fine-tuning involved challenges in understanding attention, tokenization, embeddings, training metrics, and hyperparameter selection, with debugging issues like overfitting taking up about 40% of project time. These were addressed through documentation, tutorials, experimentation, forum support, and team knowledge sharing. The main lesson learned was to start with simpler models and build a solid understanding of fundamentals before moving to complex implementations.

Model Performance Limitations

GPT-Neo 125M produced inconsistent responses, including repetition, incoherent outputs, leaked tokens, generic replies, and difficulty handling complex emotions, which affected user experience and trust. Solutions such as adjusting temperature, increasing repetition penalties, n-gram blocking, and backend response cleaning partially improved output quality. Migration to the larger Phi-2 model was considered but rejected due to slow CPU inference, high GPU costs, and cold-start delays. The key lesson learned is that small models like GPT-Neo 125M are limited in handling nuanced mental health conversations, and future versions should use larger models with GPU support for better response quality.

Firebase Integration Complexity

The main challenge was integrating Firebase with Flutter for real-time data synchronization. Difficulties included handling asynchronous operations, managing StreamBuilders efficiently, and implementing security rules correctly. Issues like “permission denied” errors and improper query results were resolved through careful testing, using async/await patterns, and optimizing widget updates. The team

also used the Firebase Emulator for local testing before deployment. This experience highlighted the importance of understanding Flutter's asynchronous model and thoroughly testing Firebase operations to ensure smooth integration.

Dataset Quality Issues

Mid-project, data problems like vague responses, inconsistent tone, typos, and occasional inappropriate content were found, leading to suboptimal model learning. The team manually cleaned key examples, applied data augmentation, and documented remaining issues. The main lesson learned was that data quality matters more than quantity, and adequate time should be spent on cleaning before training.

Hugging Face Space Downtime

The free-tier Hugging Face Space sleeps after 15–30 minutes of inactivity, causing the first request to take 30–40 seconds and leading users to think the app was broken. Workarounds included showing an informative message, increasing the app timeout, and providing fallback responses. The main lesson learned is that free-tier limitations affect user experience, and production deployment requires paid hosting or self-hosting while keeping users informed about potential delays.

4.4 Deviations from Original Objectives

This section analyzes how the final implementation compares to the original project proposal objectives.

4.4.1 Original Objectives

The project proposal specified five primary objectives:

1. To design and develop an AI system capable of engaging in empathetic, context-aware conversations.
2. To integrate voice recognition (ASR) and text-to-speech (TTS) for natural, hands-free communication.
3. To implement a safety layer that can recognize signs of emotional crisis and respond safely and responsibly.
4. To offer users supportive guidance, grounding exercises, and coping strategies without giving clinical diagnoses.

5. To ensure strong user privacy and data protection through ethical system design and secure data handling practices.

4.4.2 Negative Deviation

1. Limited Empathy and Context: GPT-Neo 125M produces inconsistent responses, repeats phrases, and cannot handle complex emotional situations or adapt to individual users.

2. No Voice Interaction: Voice recognition and text-to-speech were not implemented, reducing accessibility and natural conversation experience.

3. Basic Safety Layer: Crisis detection is keyword-based only, missing nuanced or indirect expressions and lacking escalation or integration with professional help.

4. No Structured Coping Strategies: AI provides supportive listening but cannot guide users through exercises, CBT/DBT techniques, or personalized interventions.

4.4.3 Positive Deviation

1. Functional Text-Based Conversations: Despite limitations, the AI provides context-aware and empathetic responses, helping users express emotions safely.

2. Ethical Boundaries Maintained: AI never gives medical diagnoses or replaces professional care, maintaining clear limits on its support role.

3. Strong Privacy and Security: Automatic 90-day data deletion, secure Firestore rules, user-controlled message history, and zero third-party data collection exceed original privacy goals.

4. Feasibility Demonstrated: Successfully fine-tuned and deployed an AI mental health assistant on limited hardware, showing practical real-world implementation.

Chapter 5
Conclusion and Recommendations

This final chapter summarizes the project outcomes, discusses limitations, and proposes future enhancements to transform Freud from an academic prototype into a production-ready mental health support system.

5.1 Conclusion

The Freud AI mental health companion project successfully demonstrated the feasibility of building an accessible, privacy-focused AI chatbot for emotional support using open-source tools and limited resources. Over three months, our five-member team progressed from minimal ML experience to deploying a functional system integrating fine-tuned LLMs, FastAPI backend, Firestore database, and a cross-platform Flutter mobile application. Key achievements include successful GPT-Neo 125M fine-tuning, resource-efficient deployment on free cloud platforms, privacy-first design with automatic data deletion, and cohesive full-stack integration.

Despite these accomplishments, the project has important limitations. GPT-Neo 125M produces inconsistent responses, cannot fully understand nuanced emotions, and lacks long-term memory. Crisis detection is basic and may miss subtle signs of distress, while key features such as voice interaction, structured coping strategies, and personalized guidance remain unimplemented. These limitations highlight that the system is not yet suitable for production deployment and requires further improvements for real-world use.

The project holds significant social and educational value. Freud illustrates that meaningful AI-powered mental health support is possible even with constrained resources, providing judgment-free, 24/7 assistance while safeguarding user privacy. The team gained hands-on experience across ML, cloud deployment, mobile development, and ethical considerations, learning the responsibility inherent in designing mental health technologies. Freud represents a promising first step toward democratizing mental health support and normalizing emotional well-being conversations.

5.2 Limitations

While Freud meets its core goals, certain limitations were observed:

1. Model Capacity and Response Quality: GPT-Neo 125M struggles with complex emotions, long-term context, and sometimes generates repetitive or off-topic responses.

2. Crisis Detection: Keyword-based system misses subtle distress signals and cannot escalate to professionals.

3. No Voice Interaction: Audio input/output (and) not implemented, limiting accessibility and natural conversation.

4. Limited Therapeutic Support: Lacks structured coping strategies, mindfulness exercises, or personalized interventions.

5. Infrastructure Constraints: CPU-based inference causes slow responses and cold start delays, no load balancing for high traffic.

6. Language and Accessibility: English-only interface excludes non-English speakers; no screen reader support and may not suit all cultural contexts.

5.3 Future Enhancements

There is strong potential to enhance Freud AI in future versions:

1. Larger Model for Better Responses: Upgrade to 1B+ parameter LLMs to improve understanding of complex emotional contexts, reduce repetitive or generic outputs, and handle nuanced mental health conversations.

2. Extended Context and Memory: Implement long-term memory across sessions to maintain conversation continuity, recognize recurring themes, and personalize responses based on user history.

3. Voice Interaction (and): Add text-to-speech and speech-to-text functionality to support hands-free, natural conversation and accessibility for users with visual impairments or literacy limitations.

4. Offline and Multilingual Support: Enable offline usage and expand to Nepali, Hindi, and other languages to reach non-English speakers and ensure cultural relevance.

5. Structured Coping Strategies: Integrate guided interventions like CBT/DBT techniques, mindfulness, grounding, journaling prompts, and progressive exercises to offer actionable support.

6. Advanced Crisis Detection: Move from keyword matching to ML-based risk assessment for nuanced detection, severity grading, and integration with professional resources or hotlines.

7. Personalization and Progress Tracking: Adapt conversation style, pacing, and tone to individual users, and track mood trends, coping effectiveness, and engagement metrics over time.

In summary, Freud shows that AI can provide accessible, immediate mental health support, even with limited resources. While it cannot replace human therapists, it offers a safe space for users to express emotions and practice coping skills. The project highlights both the technical and ethical challenges of mental health AI. Moving forward, careful refinement and responsible development are key to realizing its full potential.

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Appendix A

This appendix contains supplementary material that supports the main content of the report. The information provided here is intended to offer additional clarity and reference without interrupting the flow of the core chapters.

A.1 System Configuration Details

This section summarizes important configuration and setup details used during the development and deployment of the Freud AI system.

- Programming Language: Python
- Backend Framework: FastAPI
- Frontend Framework: Flutter
- Database: Firebase Firestore
- Model Architecture: GPT-Neo 125M
- Deployment Platform: Hugging Face Spaces

A.2 Hyperparameter Settings

The following hyperparameters were used during model fine-tuning. These values were selected based on experimentation and available computational resources.

- Learning Rate: 2×10^{-5}
- Batch Size: 2
- Number of Epochs: 3
- Context Window: 512 tokens
- Optimizer: AdamW

A.3 Ethical Considerations

Freud AI is designed as a supportive conversational system and not a replacement for professional mental health services. The following ethical principles were followed:

- No medical diagnosis or clinical advice is provided
- Users are encouraged to seek professional help when needed
- User privacy and data protection are prioritized
- Clear disclaimers are presented within the application

A.4 Project Timeline and Schedule

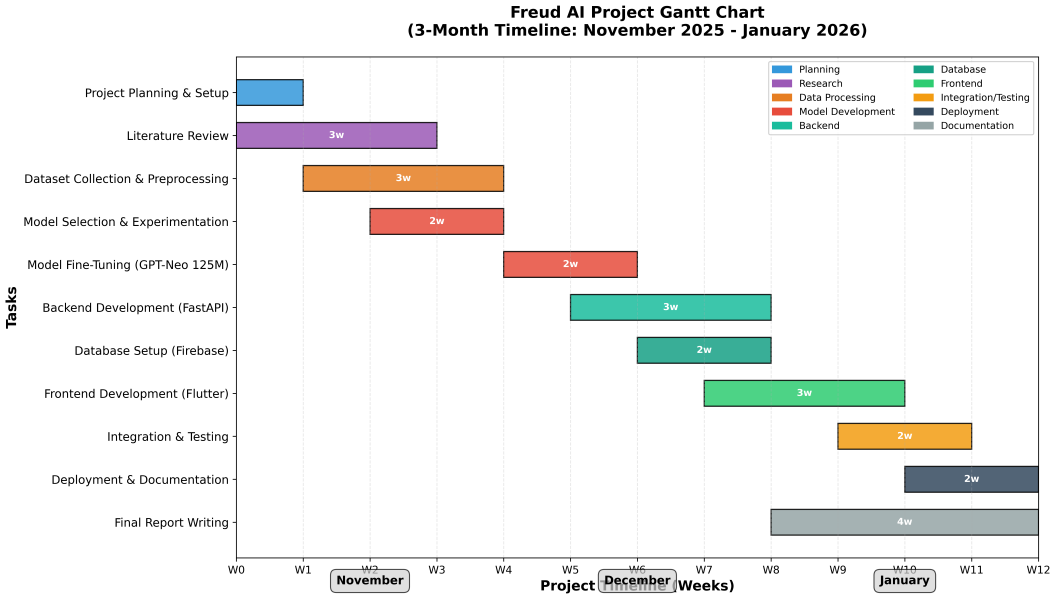

Figure 5.1: Project Gantt Chart

A.5 Screenshots

A.5.1 Database

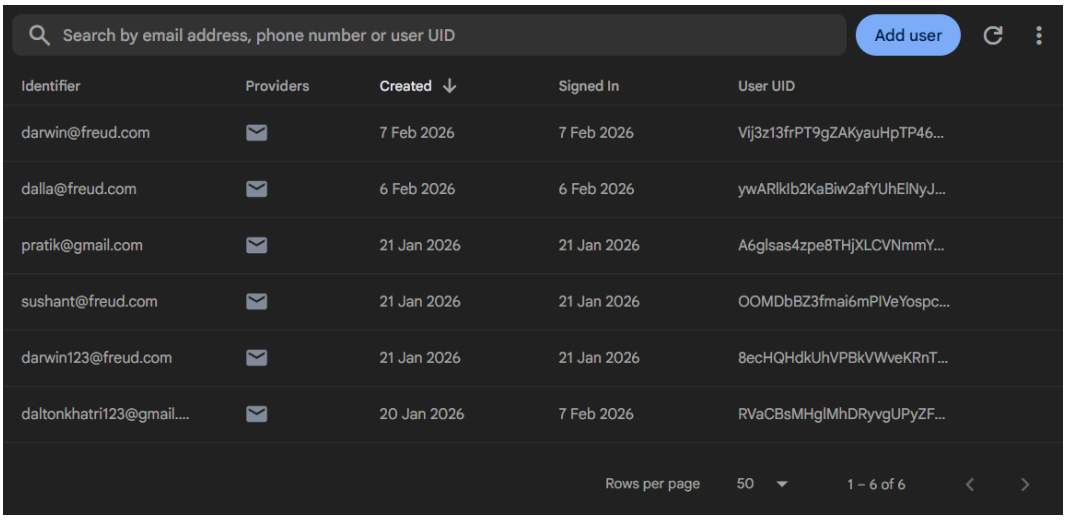

Figure 5.2: Firebase Athentication

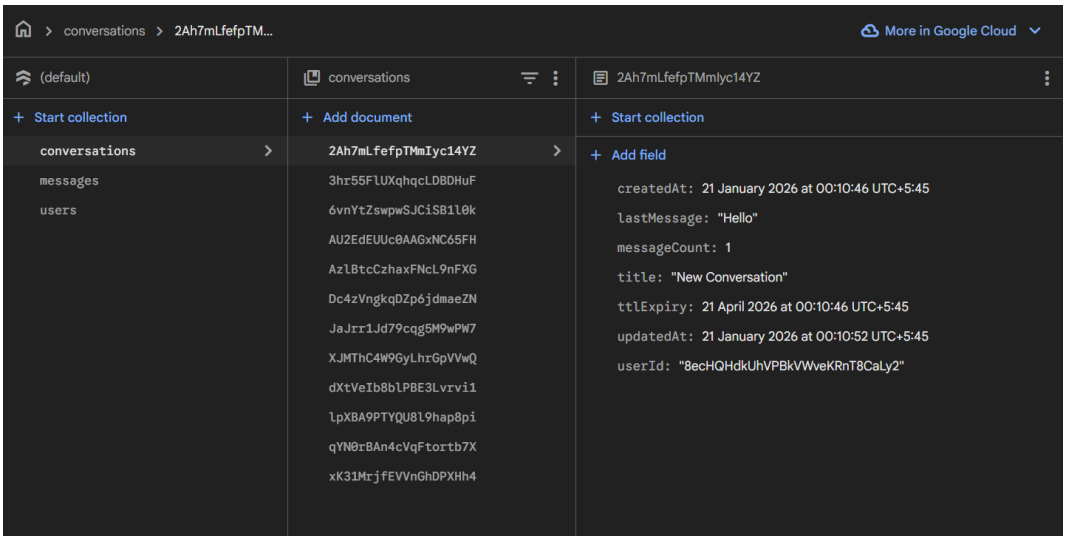

Figure 5.3: Firebase Athentication

A.6 Limitations of Supplementary Material

The appendix content reflects the system state at the time of submission. Future updates, model improvements, or infrastructure changes are not included in this section.